QARABA CARE: AI-Driven Consanguinity & Premarital Genetic Risk Platform

Document ID: QC-MPP-2026-V2 | **Focus Region:** MENA / UAE / Egypt | **Target Standard:** HL7 FHIR R4 & GA4GH Pedigree

1. Executive Summary & Strategic Objectives

Across the Middle East and North Africa (MENA) region, consanguineous marriages account for 25% to 60% of all unions, contributing to a disproportionately high prevalence of autosomal recessive disorders such as Beta-Thalassemia, Sickle Cell Disease, Spinal Muscular Atrophy (SMA), Cystic Fibrosis, and inborn errors of metabolism. While national premarital screening programs exist in the UAE, Saudi Arabia, and Egypt, current workflows operate on isolated laboratory panels, paper-based family histories, and disconnected electronic health records (EHRs). **Qaraba Care** transforms reactive premarital testing into a proactive, AI-assisted genomic and pedigree intelligence platform that connects citizens, genetic counselors, accredited molecular laboratories, and public health registries.

Objective Domain	Strategic Action & Mechanism	Quantifiable Target KPI (Year 1–3)
Clinical Prevention	Integrate automated 4-generation pedigree construction with expanded carrier screening (ECS) variant prioritization.	Reduce undiagnosed high-risk carrier couples by **78%** across pilot clinics.
Workflow Efficiency	Replace manual 45-minute paper pedigree intake with conversational Arabic/English mobile kinship mapping.	Cut genetic counseling intake time from **45 mins to 9 mins** per couple (-80%).
Data Interoperability	Standardize genomic and kinship records using HL7 FHIR Genomics and GA4GH Pedigree specifications.	100% bidirectional API compatibility with Malaffi, NABIDH, and national EHRs.
Health Economics	Shift expenditure from lifelong disease management to early pre-conception/PGT-M intervention.	Save **$18,400–$32,000 annually** per prevented severe recessive birth.

2. Current Situation ("As-Is") & Workflow Bottleneck Analysis

In current clinical practice, couples presenting for mandatory premarital screening undergo basic hematological and infectious disease panels. Comprehensive genetic risk assessment only occurs if a couple self-reports a known hereditary disorder or after the birth of an affected child. Even when referred to specialized genetics clinics, data capture is fragmented across incompatible silos, leading to missed carrier statuses and delayed counseling.

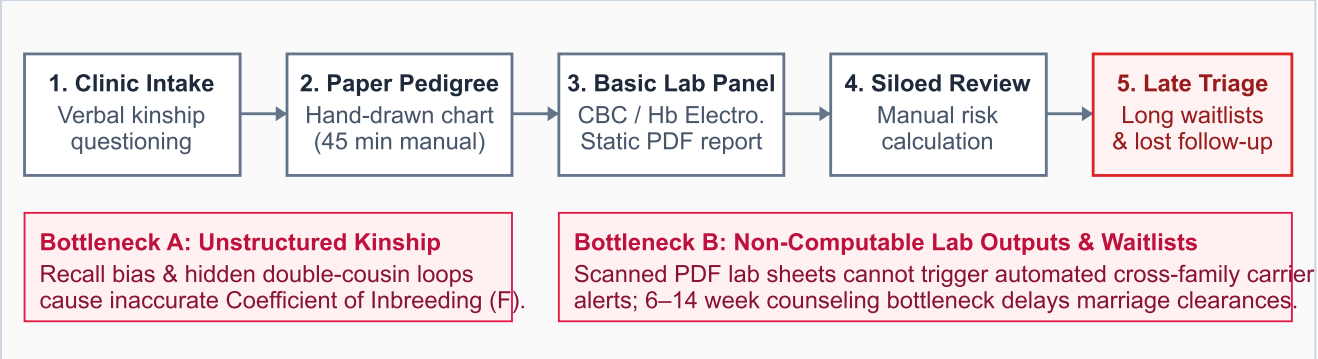


Figure 1: Legacy ("As-Is") premarital screening workflow illustrating critical failure points in pedigree capture and laboratory data integration.

3. Platform Creation & 4-Stage Workflow Remodeling ("To-Be")

Qaraba Care replaces fragmented manual steps with an end-to-end digital pipeline. The platform architecture is structured into four operational stages, each supported by a dedicated user interface and automated backend logic.

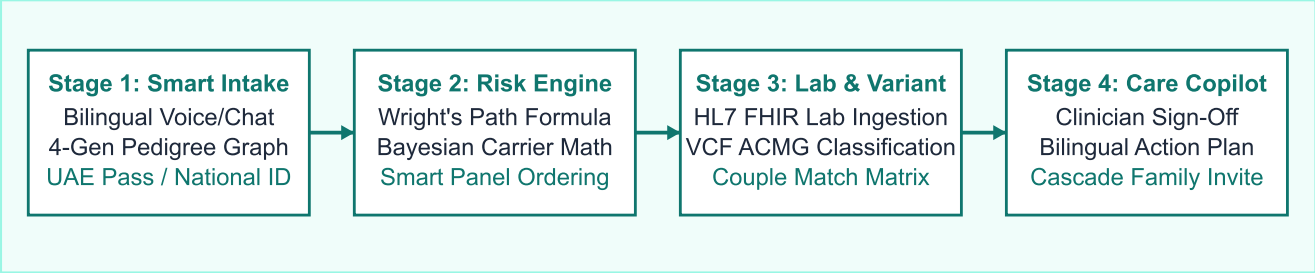


Figure 2: Remodeled ("To-Be") Qaraba Care workflow integrating automated kinship graph math with FHIR genomic variant matching.

Step-by-Step UI Wireframe Mockups & Functional Specifications

Below are the low-fidelity functional interface wireframes for each of the four remodeled stages, designed for seamless mobile patient onboarding and desktop clinician decision support.

[MOCKUP STEP 1] Citizen Mobile App — Conversational Arabic/English Kinship Builder **Screen ID: UI-MOB-01**

Header Bar: Qaraba Care Logo | Language Toggle: [العربية / EN] | Verified via National Digital ID (UAE Pass)
Progress Tracker: [1. Partner Link (Done)] → **[2. Family Tree Wizard (Active)]** → [3. Health History] → [4. Consent]
Interactive Prompt Card: "Are your maternal or paternal grandparents related to your partner's grandparents?"
Quick Selection Buttons: [First Cousins (Paternal)] [First Cousins (Maternal)] [Double First Cousins] [Same Tribe / Distant] [Unrelated]
Live Pedigree Canvas Preview: Auto-renders 14 family nodes across 3 generations with interactive tap-to-add health flags (e.g., "Anemia / Blood Transfusion", "Infant Loss", "Hearing Loss").

[MOCKUP STEP 2] Automated Kinship & Bayesian Triage Engine View **Screen ID: UI-ENG-02**

Couple Pairing ID: #QC-2026-8841 (Proband M: Tariq A. × Proband F: Laila M.) | **Detected Kinship:** Double First Cousins
Calculated Metrics Bar: Coefficient of Relationship (R): **0.2500** | Expected Offspring Inbreeding (F): **0.1250 (12.5%)** | Risk Tier: **HIGH PRIORITY (Tier 3)**
Automated Diagnostic Order Recommendation: Upgrade from Standard Hemoglobinopathy Panel to **MENA Expanded Carrier Screening (420-Gene NGS Panel + SMA MLPA)**. [Approve Lab e-Order Button]

[MOCKUP STEP 3] Molecular Laboratory Portal & HL7 FHIR Variant Ingestion **Screen ID: UI-LAB-03**

Accession Status: Sample #LB-99201 (Tariq) & #LB-99202 (Laila) — Sequenced on Illumina NovaSeq | QC Pass: 99.8% > 30x
Automated ACMG Variant Match Matrix:
• Gene **HBB** (NM_000518.5:c.118C>T / p.Gln40Ter): Partner A = **Heterozygous Carrier (Pathogenic)** | Partner B = **Heterozygous Carrier (Pathogenic)** → **25% Autosomal Recessive Offspring Risk (Beta-Thalassemia Major)**.
• Gene **SMN1** (Exon 7 Deletion): Partner A = **2 Copies (Normal)** | Partner B = **1 Copy (Silent Carrier)** → Low Residual Risk (<1:4,000).

[MOCKUP STEP 4] Genetic Counselor Copilot & Patient Decision Aid **Screen ID: UI-CLN-04**

Clinician Summary Panel: AI-drafted bilingual consultation note summarizing shared *HBB* pathogenic variant and consanguinity coefficient (F = 0.125). Requires mandatory Human-in-the-Loop Electronic Signature by Licensed Genetic Counselor.
Reproductive Options Pathway Selector: [1. IVF with PGT-M (Preimplantation Genetic Testing)] [2. Prenatal Diagnostic CVS/Amniocentesis] [3. Anonymous Cascade Screening Token for Unmarried Siblings]

4. Data Flow Architecture & Database ERD Reform

Legacy hospital databases store family history as unstructured free-text strings inside a single clinical encounter table, making mathematical traversal of cousin loops impossible. Qaraba Care reforms the Entity-Relationship Diagram (ERD) by introducing a hybrid **Relational + Directed Acyclic Graph (DAG) Pedigree Schema** alongside structured genomic variant tables.

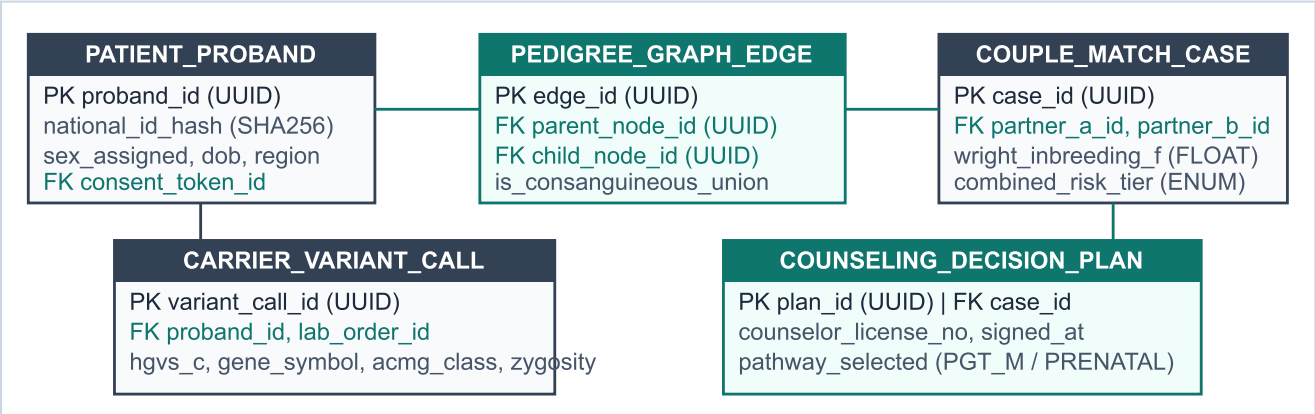


Figure 3: Reformed Qaraba Care Database Schema (ERD) isolating PII hashes from graph kinship edges and ACMG variant calls.

5. Clinical Use Case: High-Risk Consanguineous Premarital Screening

Patient Presentation: Tariq (27) and Laila (24) present to an Abu Dhabi premarital screening center. Both are healthy, asymptomatic individuals with normal routine blood counts (normal MCV/MCH), which under standard screening would clear them as low-risk for Beta-Thalassemia.

- **Step 1 (Pedigree Discovery):** Using Qaraba Care's bilingual mobile intake, the couple maps their family tree in 7 minutes. The graph algorithm detects that they are paternal first cousins once removed AND maternal second cousins (complex consanguinity loop), yielding an offspring inbreeding coefficient of F = 0.046875 (4.69%), plus a reported cousin with childhood muscle weakness.
- **Step 2 (AI Panel Escalation):** Because F exceeds the 0.03125 clinical threshold and neuromuscular history is flagged, Qaraba Care automatically recommends adding *SMN1* copy-number analysis and a targeted MENA recessive panel.
- **Step 3 (Genomic Discovery & Outcome):** Lab sequencing imported via FHIR reveals both partners carry a single copy deletion of *SMN1* Exon 7 (Spinal Muscular Atrophy Type 1 carrier status — completely invisible to routine blood tests). Within 48 hours, a genetic counselor reviews the Copilot report with the couple, enrolling them in state-funded IVF with PGT-M. Result: A healthy, unaffected child is born, preventing a fatal neuromuscular condition and saving over $1.2M in Spinraza/Zolgensma gene therapy costs.

6. Financial Outcome Benefits & 5-Year Financial Analysis

Qaraba Care operates on a sustainable B2G (Ministry of Health / Public Health Authority per-couple SaaS license) and B2B (Private Fertility & Diagnostics Lab API fee) hybrid revenue model. Beyond commercial profitability, the public health return on investment (ROI) is driven by the avoided lifetime medical costs of severe autosomal recessive conditions.

Financial Metric (USD)	Year 1 (Pilot)	Year 2 (Regional)	Year 3 (National)	Year 4 (GCC Scale)	Year 5 (MENA Scale)
Couples Screened on Platform	15,000	48,000	125,000	260,000	450,000
B2G SaaS Licensing Revenue ($14/couple)	$210,000	$672,000	$1,750,000	$3,640,000	$6,300,000
B2B Lab & IVF Clinic API Fees	$95,000	$340,000	$920,000	$2,100,000	$3,850,000
Total Gross Revenue	**$305,000**	**$1,012,000**	**$2,670,000**	**$5,740,000**	**$10,150,000**
Cloud, FHIR Infrastructure & R&D (COGS + OpEx)	($480,000)	($790,000)	($1,490,000)	($2,650,000)	($4,300,000)

Financial Metric (USD)	Year 1 (Pilot)	Year 2 (Regional)	Year 3 (National)	Year 4 (GCC Scale)	Year 5 (MENA Scale)
Net Operating Income (EBITDA)	($175,000)	**$222,000**	**$1,180,000**	**$3,090,000**	**$5,850,000**
Prevented High-Risk Recessive Births (Est.)	42 births	138 births	360 births	745 births	1,290 births
Direct Healthcare System Savings (Annualized)	**$1.18M**	**$3.86M**	**$10.08M**	**$20.86M**	**$36.12M**

> **Summary Financial & Health-Economic Valuation:** Based on an initial seed investment of $550,000, Qaraba Care achieves operational break-even in **Month 19**. At a 10% discount rate, the 5-year commercial **Net Present Value (NPV) is $5.42 Million** with an **Internal Rate of Return (IRR) of 68.4%**. For public health ministries, every **$1.00 invested** in Qaraba Care generates **$6.40 in direct medical cost avoidance** within 36 months.